Pallet + Parts Tracker

Features and uses

A guide for everyone who works with the shop-floor tracker: what it keeps track of, how to do the everyday jobs in it, and what each screen is for.

Built by **HiFen Kong**, Metal Fab, Van 1.

> **In one sentence:** the tracker answers "where is this job's work, what state is it in, and when is it due?" from a phone or computer, so nobody has to walk the floor or ask around.

1. What the app is for

The tracker replaces the whiteboard, the sticky notes, and the "has anyone seen the..." questions. It keeps one shared record of every job, the parts in it, the pallets those parts sit on, and where each pallet is on the floor. Everyone sees the same thing, updated as people work.

Questions it answers quickly

- Which jobs are not shipped yet, and which are overdue or due soon?
- What stage is this part at: cut, threaded, bent, complete?
- Which pallet is it on, and which zone is that pallet in?
- What is on this pallet, and for mixed pallets, which job does each part belong to?
- How many pallets does this job have, and how many have shipped?
- Who moved or changed something, and when?
- When was a job completed, wrapped, and shipped?

Who it is for

Everyone who has a login has the same access. There are no separate manager screens, so shop staff, shipping, and office staff all use the same app. Every change is recorded with the name of the person who made it.

2. The key ideas

A few terms are used throughout the app.

Term	What it means
Job	A piece of work, identified by its job name or number. Job numbers do not have to be unique; the line (below) and the project help tell jobs apart.
Line	An optional label on a job, for example 2, 1-5, or 500/501. Big jobs have lines; one-part jobs do not. When several spreadsheet lines are made together, you type one combined label such as 500-505.
One-part job	A job with no line and a single part. Its part number and quantity come from the spreadsheet and it is tracked from start to finish.
Big job	A job with a line and a total quantity but no part list yet. Its parts are entered as they are put on pallets.
Part	What is being made: a part number (it can be blank), a quantity, and a stage.
Stage	Where a part is in the process: Waiting to cut, Cut, Threading, Bending, Notching, Deburring, Complete. Any stage can be chosen at any time.
Job pallet	A pallet holding parts from one job. Named after the job and numbered, for example CDP5000 #1, CDP5000 #2.

Term	What it means
Mixed pallet	A pallet holding parts from different jobs. You give it a name, and pallets with the same name are numbered #1, #2, and so on.
Zone	A named area of the floor where pallets are kept. Zones can carry notes and can be drawn on the floor map.
Pallet status	Open, wrapped, or shipped. The date each is reached is recorded.

3. Getting started

1. Open the address your tracker computer gives out, for example **http://192.168.1.50:5000**, on a phone, tablet, or computer connected to the shop wifi. Bookmark it. The window the tracker runs in and the **Connect** link at the bottom of the app both show the current address.
2. The very first time, the app asks you to create the first login. After that, log in with your own name and password.
3. Use the links at the top: **Add job**, **Add part**, **Pallets**, **Zones**, **Map**, **Shipped**. The links at the bottom of every page are **Activity log**, **Users**, **Export**, **Backups**, and **Connect**.

Things that are always true

- **Names are in capitals.** User names, part numbers, zone names, and pallet names are stored in all capitals, and the boxes show capitals as you type. Typing lower case never causes a mismatch.
- **Big buttons.** The pages are built for phones and gloved fingers, with large buttons and drop-downs.
- **Warnings, not roadblocks.** If you record more than a job's quantity, the app warns you but lets you carry on, because reality is sometimes ahead of the paperwork.
- **Delete asks first.** Every delete shows exactly what will go and needs a second tap to confirm.

4. Everyday workflows

A. A new one-part job comes in

1. Tap **Add part**. Type the job name, leave the line blank, and tap Next.
2. If the job does not exist yet, the app offers to create it. Enter the part number, quantity, and starting stage, add a due date or project if you have them, and save.
3. As the part moves through the shop, open the job and change its stage in the drop-down. It saves as soon as you pick it.

B. A big job with lines

1. Tap **Add job**. Enter the job, the line label you want (for example 500-505), and the total quantity for all of its parts. Add a due date and project if you have them.
2. No parts are entered yet. They are added when they go on a pallet (workflow C).
3. When every part is on a pallet, open the job and tap **Mark job complete**.

C. Putting parts on a pallet

1. Open **Pallets** and tap **New single-job pallet** (pick the job) or **New mixed pallet** (type a name). A zone is optional and can be added later.
2. On the pallet, tap **Add parts to this pallet**. For a mixed pallet, choose the job first.
3. Pick the part and the quantity. Or choose *a part not listed yet* and enter its part number and whether it is straight or bent. This is how a big job's parts get created.
4. **Add all** puts everything left of the job on the pallet in one tap.
5. A part becomes Complete automatically once its full quantity is on pallets, so you never have to click through stages that already happened.

D. Moving things around

1. On a pallet, tap **Move / take off** beside a part.
2. Enter how many, then choose *Take off* (used up or taken away), *A new pallet for this job*, or another pallet. A job pallet will only list pallets that can accept the part: mixed pallets and pallets of the same job.
3. Parts taken off are remembered ("3 taken off") so they are not counted as missing.

E. Wrapping and shipping

1. Open the pallet and change its **Status** to wrapped, then shipped. The date is recorded each time.
2. When every pallet of a job is shipped, the job leaves the main list and moves to **Shipped**, and the app tells you how many pallets were shipped. A big job also needs its full quantity on pallets, or to have been marked complete, so shipping its first pallet does not hide it.

F. Finding something

1. Type a job name, part number, or pallet name into the search box on the home page.
2. Part results show the stage, which pallet it is on, and that pallet's zone. Tap a pallet to see everything on it.
3. On a pallet's page, **Show on the floor map** flashes its zone on the map.

5. Screen-by-screen guide

Every screen in the app, and what it is for.

Screen	What it shows and what you can do there
Jobs (home)	Every job not yet shipped, with part, quantity, pallets, status, and due date. Click a column heading to sort. Overdue jobs are shaded red and jobs due within 7 days amber. Filters show only overdue jobs, or overdue and due soon. A search box sits at the top.
Job page	The job's details, status, and dates (complete, wrapped, shipped). Its parts, each with a stage drop-down and an Edit link. Its pallets. Buttons for Summary, Edit, Delete, and History. Add a part or a pallet right there.

Screen	What it shows and what you can do there
Add job / Add part	Add part can create the job at the same time, for one-part jobs. If matching jobs already exist, it lists them so you can pick the right one.
Pallets	Every pallet not yet shipped, sortable by name, zone, status, and quantity. Buttons to create job pallets and mixed pallets, and to see all pallets by group.
Pallet page	Zone and status drop-downs, what is on the pallet, Move / take off, Add parts, rename (mixed pallets), Delete, and History.
All pallets by group	Every pallet grouping on one page with its pallet count. Each pallet lists its parts and quantities, and mixed pallets show each part's job. Can include shipped pallets and can be printed.
Zones	A list of zones with their notes and the pallets in each. Add, rename, add notes, or delete zones. A zone's own page lists its pallets with their contents.
Map	The floor drawn as zones, walls, racks, and machines you can move and resize. Tap a zone to see its pallets. See section 7.
Shipped	Every shipped job with how many pallets were shipped and when it was completed, wrapped, and shipped. Sortable.
Job summary	A one-page printable sheet for a job: dates, due note, quantities (on pallets, taken off, not yet on a pallet), its parts, and its pallets.
Activity log	A record of every change: when, who, and what. Searchable and filterable by person.
Users	Add people, rename them, reset passwords, and delete accounts.
Export	Downloads jobs, parts, pallets, and the activity log as spreadsheet files.
Backups	Make a backup now, download backups, and see where they are kept.
Connect	The address to type into phones and tablets, with tips if it stops working. Open it on the tracker computer. Can be printed.

6. Uploading the job spreadsheet

Instead of typing every job, upload the spreadsheet (Excel or CSV) from the home page. Nothing is saved until you have checked it.

1. **Upload** the file. The first row must be the column headings.
2. **Check the columns.** The app guesses which column is the job, line, quantity, part number, due date, project, and assigned date. Fix any wrong guesses and say how text dates are written (year-month-day, month/day/year, or day/month/year).
3. **Combine lines.** For a job with several lines, give lines made together the same group and type the label you want, such as 500-505. Lines left alone stay separate jobs. A split such as 500/501 and 502-505 is simply two groups.
4. **Review and apply.** The review table shows what will happen to every job.

Result	What it means
New job	Nothing like it exists yet, so it is created. A row with a line becomes a big job with its total quantity. A row without a line becomes a one-part job.
Changed	A matching job exists but its quantity, due date, project, or assigned date differs. Applying creates a **new copy** with the new values. The old job is kept untouched.
No change	An existing job already has these values. Uploading the same file twice never piles up copies.
Skipped	Jobs that only exist as shipped jobs, repeats within the file, and rows without a job or (for one-part jobs) a quantity. Skipped rows are listed with the reason.

7. The floor map

The map is a rough plan of the shop floor. Each zone is a rectangle showing its name and how many pallets are in it. Zones holding pallets are shaded green.

- **Viewing:** tap a zone to open its page and see exactly which pallets and parts are there.
- **Laying it out:** tap **Edit layout**. Drag a zone to move it and drag its blue corner to resize it. Changes snap to a grid and save the moment you let go. Editing is a separate mode so nobody moves a zone by accident.
- **New zones:** a zone that is not on the map yet has a **Place on map** button. **Remove from map** takes a zone off without deleting it.
- **Walls, racks, and machines:** in edit mode, add them from the form under the map, give racks and machines a name, drag them into place, resize them, rotate them, or remove them. They are only a picture to help you read the map: they never block anything, and they can sit on top of zones.
- **Finding a pallet:** from a pallet's page, **Show on the floor map** flashes its zone.

8. Tools for managers and coordinators

Open to everyone, not limited to managers.

Due dates

Overdue and due-soon jobs stand out on the home page, with a note such as "overdue by 3 days" or "due in 2 days". The counts are shown above the list, and the filters narrow it to what needs attention.

History of anything

The **History** buttons on jobs, parts, pallets, and users open the activity log filtered to that one thing. A job's history includes its parts going on and off pallets. A user's history shows what they did and any changes made to their account. Part and account history start from the version that introduced them; earlier entries still appear in job, pallet, and person histories.

Counts and dates

Each job shows how many pallets it has. Each mixed pallet group and job pallet group shows its pallet count on the **All pallets by group** page. Completed, wrapped, and shipped dates are filled in automatically for jobs, parts, and pallets.

Exporting

File	Contents
Jobs	Every job, shipped or not: quantity, pallets, status, due date, and the dates it was completed, wrapped, and shipped.
Parts	Every part with its stage, and how many are on pallets, taken off, or not yet on a pallet.
Pallets	One row per part on each pallet, with the zone and status. Mixed pallets include the job.
Activity log	Everything recorded, oldest first.

Accounts

The **Users** page adds people, renames them, resets passwords, and deletes accounts. You cannot delete the account you are using or the last remaining account. Past log entries keep the name that was used at the time.

9. Keeping the data safe

Backups

- The first visit each day makes a backup automatically, and the newest 60 are kept.
- On the **Backups** page, tap **Back up now** at any time and download any saved backup.
- Backups kept on the same computer do not help if that computer fails. Either download one now and then to a USB stick or another computer, or set a folder (USB drive, shared folder, OneDrive) that every backup is copied to automatically.
- Restoring is a few steps with the app stopped, described in the HOSTING guide that comes with the app.

Where it runs

The app runs on one computer in the building that stays on. Start it by double-clicking **start-tracker.bat** (Windows) and leave its window open. Phones and other computers on the shop wifi connect to that computer's address. The address is shown in the tracker's window and on the **Connect** page. Changing the router is not needed: if a bookmark ever stops working, check the current address there. To keep the address permanent, ask whoever looks after the building's network to reserve it for the computer. Also stop the computer from sleeping and set it to start the app when it turns on. All the steps are in HOSTING.md.

Moving to the internet later

The app is built so it can move to a hosting service later. Before it goes on the open internet, protection against forged form submissions and limits on login attempts should be added. That is not needed while it stays on the shop wifi.

> **Security basics on the shop wifi:** use a password-protected wifi network, do not share the app's address outside the building, and never set up port forwarding on the router to reach it from the internet.

10. Rules the app follows

Situation	What the app does
A part's full quantity is on pallets	The part is marked Complete and its complete-on date is set.
A one-part job has parts on pallets	Its status shows the earliest status among its pallets: On pallet, Wrapped, or Shipped.
A big job	Status shows progress such as "40 of 120 on pallets" (and any taken off), and Complete once you mark it.
Every pallet of a job is shipped	The job is shipped: it leaves the main list and appears under Shipped with its pallet count.
A new pallet is created	It takes the lowest unused number under its name, so the number of a deleted pallet is reused.
A part is taken off a pallet	Its taken-off quantity goes up, and it counts as dealt with rather than missing.
More is recorded than the job holds	A warning is shown, and the change is allowed.
A spreadsheet re-upload has changes	A new copy of the job is created. The old job is never overwritten.
Anything is added, changed, or deleted	It is recorded in the activity log with the person's name.

11. Good to know

- Moving or resizing a zone on the map is not written to the activity log, to keep the log readable.
- Remakes and extra parts set aside are not tracked yet.
- Job and project names keep the capitalisation you type; user, part, zone, and pallet names are always capitals.
- Two jobs can share a name, for example after a re-upload creates a copy. Quantity, due date, and status tell them apart, and the old one can be deleted.
- Everything is stored in one file inside the app's **instance** folder. Keep that folder when updating the app.